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MILITARY SPECIFICATION

POWER SUPPLY PP-2581C/A

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, manufacture and acceptance requirements for the Power Supply, PP-2581C/A, for use in the Guided Missile Launcher LAU-7 Series (MIL-L-22620) to provide direct current (DC) power, firing circuit power, audio amplification, feedthrough wiring and passage of high pressure gas.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-27	Transformers and Inductor (Audio, Power, and High Power Pulse), General Specifications for
MIL-P-116	Preservation, Methods of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1440

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DOD-D-1000	Drawings, Engineering and Associated Lists
MIL-B-5087	Bonding, Electrical, Lightning Protection for Aerospace Systems
MIL-I-46058	Insulating Compound, Electrical (Coating Printed Wiring Assemblies)
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-R-6106	Relays, Electromagnetic, (Including Established Reliabiliity (ER) types) General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-18303	Test Procedures, Reproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-I-22129	Insulation Tubing, Electrical, Polytetrafluoroethylene Resin, Nonrigid
MIL-L-22620	Launcher, Guided Missile, LAU-7 Series
MIL-I-23053	Insulation Sleeving, Electrical, Heat Shrinkable, General Specification for
MIL-M-38510	Microcircuit, General Specification for
MIL-C-39006	Capacitors, Fixed, Electrolytic, (Nonsolid Electrolyte), Tantalum, Established Reliability, General Specification for
MIL-I-81219	Indicator, Elapsed Time, Electrochemical
MIL-C-85485	Cable, Electric, Filter Line, Radio Frequency Absorptive

STANDARDS

FEDERAL

FED-STD-H28	Screw-Thread Standards for Federal Services

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-198	Capacitor, Selection and Use of
MIL-STD-199	Resistor, Selection and Use of
MIL-STD-202	Test Methods for Electronic and Electrical Parts
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics; Measurement of
DOD-STD-480	Configuration Control Engineering Changes, Deviations, and Waivers

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MIL-STD-701	List of Standard Semiconductor Devices
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-726	Packing Requirements Code
MIL-STD-750	Test Methods for Semiconductor Devices
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1346	Relay, Selection, and Application
DOD-STD-1399/300	Interface Standard for Shipboard Systems, Section 300 Electric Power, Alternating Current (Metric)
MIL-STD-1562	Lists of Standard Microcircuits
MIL-STD-1568	Materials and Processes for Corrosion Prevention and Control in Aerospace Weapons Systems
MIL-STD-1629	Procedures for Performing a Failure Mode, Effects, and Criticality Analysis
MIL-STD-2068	Reliability Development Tests
MIL-STD-2074	Failure Classification for Reliability Testing

HANDBOOKS

MILITARY

MIL-HDBK-241	Design Guide for Electromagnetic Interference (EMI) Reduction in Power Supplies
MIL-HDBK-217	Reliability Prediction of Electronic Equipment

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DOCUMENTS

AS-4613	Application and Derating Requirements for Electronic Components, General Specifications for
NAVAIR 11-75A-54	Technical Manual for LAU-7 Guided Missile Launcher
WS-6536	Procedures and requirements for preparation and soldering of electrical connections
LC-78-1	Storage Reliability of Missile Material Program, Missile Material Reliability Prediction Handbook, Parts Count Prediction

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

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30003-60A89C350	Cover, Protective
30003-60A89D700	Power Supply PP-2581 C/A
30003-58A164H598	Housing Launcher Assembly

NAVSEA

10001-1517259	Connector, Plug
10001-1517260	Connector, Receptacle
10001-1517261	Connector, Receptacle
10001-59A44B303	Valve, Relief

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specifications and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Power supplies furnished under this specification shall be products which are qualified for listing in the applicable Qualified Products List (QPL) at the time set for opening of bids (see 4.4 and 6.3).

3.2 Selection and quality verification of parts, materials and processes. Quality verified parts and materials shall be used in the fabrication of the power supply. Parts and materials shall be procured from QPL sources. The contractor shall prepare test reports to conform to applicable requirements. Derating of components in accordance with the guidelines in AS-4613, Class A, shall be made to ensure reliable operation under environmental conditions. All noncompliance with derating criteria shall require written approval by the procuring activity prior to acceptance. The derating shall be such that the maximum junction temperature of semiconductor devices shall not exceed 110o Centigrade (C).

3.2.1 Stress analysis. The contractor shall determine the stress applied to each part to verify that all derating requirements are met under worst case environments, worst case continuously applied power (external to the power supply) and nominal part and circuit parameters (see 6.2.2).

3.2.2 Prohibited application. The use of electron tubes, fixed paper and plastic dielectric capacitors in molded cases, plastic or epoxy, or other encapsulated semiconductors, and microelectronic devices, germanium semiconductors and photo couplers (photo transistors) shall be prohibited.

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3.2.3 Restricted application. The use of nonsolid sintered-slug tantalum capacitors shall be restricted to MIL-C-39006, CLR 79 and CLR 80 qualified devices. Aluminum electrolytic capacitors shall require approval of the requiring activitiy.

3.2.4 Nonstandard parts and material approval. Approval for the use of nonstandard parts and materials, including transistors and diodes, other than microelectronic devices, shall be obtained as specified in MIL-E-5400 and MIL-STD-965. The contractor shall prepare requests for approval of nonstandard parts and proposed addition to an approved Program Parts Selection List (PPSL) (see 6.2.2).

3.2.5 Parts. Each electrical component part shall be tested as specified in 4.5.2 for conformance to its requirements. Lot sampling of parts is not acceptable.

3.2.5.1 Microelectronic devices. Microelectronic devices shall meet the requirements of MIL-M-38510 Class B as a minimum and shall be selected and applied in accordance with MIL-STD-454, Requirement 64.

3.2.5.2 Firing circuit relays. The firing circuit relays shall be interconnected as shown in Figure 9. Relay selection shall be in accordance with MIL-STD-1346 and MIL-R-6106. Electrical/mechanical relays shall be used. Arc suppression shall be applied to all relays. Contact gaps shall be not less than 0.005 inch. Minimum contact ratings are as follows:

Relay 1/	Rating
K5, K6	1.0 Amperes DC at 29 V DC Resistive Load
K2, K3	10.0 Amperes DC at 29 V DC Resistive Load
K4	10.0 Amperes DC at 29 V DC Resistive Load 2.0 Amperes at 115 V AC Resistive Load 2.0 Amperes at 29 V DC Inductive Load 0.5 Amperes at 265 V DC Resistive Load

1/ Abbreviated designations only. Complete designations to be developed by the manufacturer.

Operating time for each relay (time to close) shall be not greater than 20 milliseconds (ms) under standard conditions (See 3.3.2.3) at 25.6 V DC coil voltage. Relay pull-in voltage (common contact transfer) shall be no greater than 18V DC under all service (See 3.5) conditions. Relay dropout voltage shall be 1.5 to 10.0 V DC throughout the temperature range of -55 to +85°C. Relay Life shall not be less than 50,000 cycles comprising one make, one break of each set of contacts under above rated load conditions. Relay contacts shall not transfer or chatter except as specified in MIL-R-6106 when subjected to sinusoidal and random vibration as follows:

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a.	Vibration-sinusoidal	6 pole	2 and 4 pole
	G-level	20 G	30 G
	Frequency range	10-2,000Hz	10-2,000 Hz
b.	Vibration-random	6 pole	2 and 4 pole
	Power spectral density	0.3 G^2/Hz	0.4 G^2/Hz
	Frequency range	50-2,000 Hz	50-2,000 Hz

3.2.5.3 Semiconductor devices. Semiconductor devices used in the power supply shall be selected and applied in accordance with MIL-STD-701 and MIL-S-19500. No plastic-encapsulated semiconductors shall be used. Semiconductor devices shall be class JANTX, or procured to the same product assurance level as JANTX (See 3.11).

3.2.5.4 Transformers and inductors. Transformers and inductors used in the power supply shall be manufactured in accordance with the applicable paragraphs of MIL-T-27 insulation class S and shall meet Group B inspection.

3.2.5.6 Capacitors. Capacitors in the power supply shall be selected and applied in accordance with MIL-STD-198 failure level P as a minimum.

3.2.5.7 Resistors. Resistors in the power supply shall be selected and applied in accordance with MIL-STD-199 failure level P as a minimum.

3.2.6 Materials and processes. The selection of materials and processes for the power supply shall be in accordance with MIL-E-5400, MIL-STD-454, and MIL-STD-1568. Appropriate procedures shall be observed during handling of devices susceptable to electrostatic discharge damage. Selection of materials shall be made with special attention being given to the requirements of temperature, temperature change, thermal conductivity, vibration, loads, other environmental conditions, reliability, and lightweight construction.

3.2.6.1 Screw threads. Screws, nuts, bolts, and threaded parts shall be in accordance with FED-STD-H28.

3.2.6.2 Multiconductor cable. Multiconductor cable shall be in accordance with MIL-STD-454, Requirement 66.

3.2.6.3 Electrical filters. Electrical filters shall be in accordance with MIL-STD-454, Requirement 70.

3.2.6.4 Unshielded insulated wire. Unshielded insulated wire shall meet the requirements of MIL-C-85485. Conductors shall be 20 AWG minimum stranded wire except Ground Ring connections shall be 16 AWG minimum stranded wire (see 3.3.1.1).

3.2.6.5 Shielded wire. Shielded wire shall meet the requirements of MIL-C-85485. Center conductors shall be 20 AWG minimum stranded wire. Pig-tail shield terminations shall not be used.

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3.2.6.6 Shielded wire braid. The braid on shielded wire shall use a heat shrinkable insulating sleeving containing a solder ring with flux and meltable sealing rings for connections to ground wires. Compression type connections which may cause conductor teflon insulation to cold flow during service use shall not be used.

3.2.6.7 Heat shrinkable insulation sleeving. Heat shrinkable insulation sleeving conforming to MIL-I-23053 shall be used over axial lead wire termination and at locations where additional insulation is required to prevent contact of adjacent electrical connections.

3.2.6.8 Teflon insulated sleeving. Teflon insulation sleeving conforming to MIL-I-22129 shall be used to protect component lead wires from contacting adjacent leads or components.

3.2.6.9 Insulators, insulating material and dielectric material. Insulators, insulating material, and dieletric material shall be in accordance with MIL-STD-454 Requirement 11.

3.2.6.10 Conformal coatings, encapsulants, embedments, and potting materials. Conformal coatings, encapsulants, embedments, and potting materials if used with modular assemblies containing microelectronic devices and discrete parts shall be compatible with the power supply encapsulants if used and shall be in accordance with MIL-STD-454, Requirement 47. Conformal coating of printed wiring boards is required and shall conform to MIL-I-46058.

3.2.6.11 Corrosion resistance. Ferrous alloys shall be selected and applied in accordance with MIL-STD-454, Requirement 15.

3.2.6.12 Dissimilar metals. Dissimilar metals shall not be used in contact with each other unless protected against electrolytic corrosion. Classification and use of dissimilar metals shall be in accordance with the provisions of MIL-STD-454, Requirement 16.

3.2.6.13 Fungus resistance. Materials that are nutrient for fungi shall not be used.

3.2.6.14 Deleterious fumes. Use of materials that will liberate deleterious fumes under any environmental condition that may be experienced in testing, storage, or service usage is prohibited.

3.2.6.15 Finish. All aluminum exterior surfaces shall be finished in accordance with anodic coating in accordance with MIL-A-8625. Interior surfaces shall be finished in accordance with MIL-E-5400.

3.2.6.16 Wiring. Internal wiring practices shall be in accordance with MIL-STD-454, Requirement 69. In addition, teflon type insulation and sleeving, shall be etched to insure adhesion to potting materials (See 3.2.6.10).

3.2.6.17 Soldering. Soldering shall be in accordance with WS-6536.

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3.2.6.18 Corona and electrical breakdown prevention. Corona and electrical breakdown prevention shall be in accordance with MIL-STD-454, Requirement 45.

3.2.6.19 Safety (personnel hazard). Provisions for personnel safety shall be in accordance with MIL-STD-454, Requirement 1.

3.2.6.20 Grounding and bonding. Grounding and bonding shall be in accordance with MIL-B-5087. Only one chassis ground shall be allowed (See 3.3.1.1).

3.2.6.21 Printed wiring. Printed wiring shall be in accordance with MIL-STD-454, Requirement 17.

3.3 Construction and design. The power supply shall meet the following construction and design requirements specified in 3.3.1 through 3.3.1.5.

3.3.1 Power supply construction. Construction shall be such that physical and functional relationships between power supply components shall not be degraded by strains, impacts, and vibrations associated with operation, storage, and shipping.

3.3.1.1 Power supply wiring. The power supply shall be wired to provide the interconnections specified in TABLE I and shown on Figure 9. Any other design shall require the approval of the procuring activity.

3.3.1.2 Form factor and serialization. The power supply shall conform to the drawings specified in 30003-60A89D700. Power supply serial numbers shall be in accordance with MIL-N-18307 (See 6.2.2).

3.3.1.3 Connectors. Connector receptacles J1 and J2 shall meet the requirements of Drawings 10001-1517261 and 10001-1517260 respectively and shall mate and operate with connector plugs defined by Drawing 10001-1517259. The wire termination side of the connector shall be potted with an epoxy type compound.

3.3.1.4 Connector cover. A reusable metal protective cover (See Drawing 30003-60A89C350) shall be installed on connector receptacle J2. The cover shall provide a water-tight seal on the receptacle. The cover shall lock on the receptacle with detent action. A plastic dust cap shall be provided on receptacle J1 for protection during shipping.

3.3.1.5 Weight. The power supply weight shall be 10.5 (±1.5) pounds (lbs) with the center of gravity within plus or minus two inches of the geometric center of the power supply. The design goal is 10.5 lbs.

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TABLE I. Feedthrough wiring.

From J1-	To J2-	Cable 2/	Notes
7 1/	1	3	400 Hz Test
19	4	1	N2 Valve Ground
20 1/	5	1	ACQ Signal (LAMBDA)
10 1/	7	2	ACQ number 2
3 1/	16	2	ACQ number 1
27 1/	10	1	Spare
26	14	1	Power Interlock Ground
13	17	1	Missile Identification
12 1/	26	2	U-D
1 1/	28	2	TM ON-OFF
22 1/	34	1	R-L
17 1/	8	2	Spare
14 1/	24	1	Spare
35	35		N2 Line
	11		N2 Relief Valve

1/Indicates shielded wire. Shield is grounded at J1.
2/Cable numbers 1, 2, and 3 indicate grouping of feedthrough wires. The separation between each group shall not be less than 0.2 inch.

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3.3.2 Power supply design. The power supply shall meet the design requirements specified in 3.3.2.1 through 3.3.2.9. In addition, the power supply shall suffer no damage during power on starts, power switching and dropouts, surges, and load changes. The power supply shall not be damaged by power as stated in MIL-STD-1399/300.

3.3.2.1 Cooling. The power supply shall require no cooling.

3.3.2.2 Thermal design. The power density (output power of the high and low voltage supplies divided by the volume occupied by the high and low voltage supplies) of the power supply shall be not greater than 3 watts per cubic inch. Junction temperature for semiconductor devices shall be not greater than 110° C. Case or hot spot temperatures for all other electronic devices shall be not greater than 40° C rise above power supply chassis ambient with a maximum absolute temperature of +110° C for devices less than or equal to 3 watts; 55° C rise above power supply chassis ambient with a maximum absolute temperature of +125° C for devices greater than 3 watts; 30° C rise above power supply chassis ambient with a maximum absolute temperature of +100° C for transformers, and +10° C rise above power supply chassis ambient, due to self heating for capacitors.

3.3.2.3 Standard Conditions. The standard inputs, loads, and environmental conditions shall be as follows unless otherwise stated:

a. Temperature - room ambient

b. Altitude - normal ground level

c. Vibration - none

d. Humidity - Room ambient up to 90 percent relative humidity

e. Input power
   1. AC - 115(±1.0)VAC, 400(±4)Hz, 1 phase
   2. DC - 28(±0.5)VDC

f. Supply loads (FULL)
   1. Low voltage - 1.5(±10 percent) amps
   2. High voltage - 175(±10 percent) ma

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g. Audio Amplifier
   1. Input - 10V RMS, 1500 Hz
   2. Output - 600(±1 percent) ohm load

h. Relay - Relay loads are as specified in 3.4.5

3.3.2.4 Electrical overload protection. Overload protection for the power supply shall be provided internal to the power supply. All parts and circuits which are likely to carry an overload due to any failures or poor adjustments shall be proportioned to withstand such overload without permanent damage.

3.3.2.5 Undervoltage protection. The power supply shall not be damaged by voltages below the minimums specified in 3.4 and shall automatically resume operation when the voltage returns within normal limits.

3.3.2.6 Firing circuit safety. Relay outputs shall be less than 0.0001 amps at all "D" conditions of TABLE IV at any condition of input power, load or load transition and before, during and after relay energize and deenergize. Firing circuit relays K2, K3, and K4 (See FIGURE 9) shall be hard mounted, hard wired and the headers potted. Spacing shall be maintained to prevent leakage from any other adjacent wiring or components which could energize the firing circuit.

3.3.2.7 Output switching. Relay switched outputs of the high-voltage and low-voltage supplies shall not be permitted.

3.3.2.8 Relay coil transients. Relay coil transients shall be suppressed to a maximum of 42 volts across relay coil terminals.

3.3.2.9 Failure containment requirements. Any electrical failure of the power supply shall be contained within the power supply. In addition, the manufacturer shall prepare a Failure Mode, Effects, and Criticality Analysis in accordance with MIL-STD-1629, Task 101 (see 6.2.2).

3.4 Performance. The power supply shall provide performance specified herein, under standard and service conditions when energized from power sources having characteristics and limits as defined in MIL-STD-704 except normal limits are:
   a. AC power (single phase), 98-124 VAC, 380-420 Hz.
   b. DC power, 18-30 VDC.

3.4.1 Insulation resistance. With a high potential of 500 VDC insulation resistance shall be 100 megohms minimum between each isolated circuit and all other circuits, including ground (See 4.4.2.2). Circuits which will be damaged by this test will not be subjected to this test.

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3.4.2 Gas system. A feedthrough gas system shall be provided to allow the passage of dry air or nitrogen for missile cooling. The gas will be at 3000 psig working pressure with a peak pressure of 4700 psig at service conditions. The gas system shall have a maximum leakage rate of 1 cubic centimeter per minute at a pressure of 3800 (± 50) psig and 2200 (±50) psig at 25 (± 5)o C. The pressure drop across the feedthrough gas system shall be not greater than 50 psig when the inlet at J1 is maintained at a pressure of 3000 (±100) psig at a flow rate of 28.3 standard liters per minute (1 cubic foot per minute). The cleanliness of the gas system shall be as detailed herein. The power supply gas system shall be visually examined for cleanliness and shall be free of all water, oils, grease, and foreign matter exceeding 0.002 inches in size on surfaces internally interfaced with the nitrogen (gas) system. Pin 11 of J2 shall have the tube connected to a pressure relief valve (Drawing 10001-59A44B303) as illustrated in Drawing 30003-60A89D700. Cracking pressure of the pressure relief valve shall be from 15-40 psig. Reseat pressure of the pressure relief valve shall be 10 psig minimum. This valve shall be equipped with a deflector cap (See 4.4.2.3).

3.4.3 High and low voltage DC. The power supply shall provide electrical power for the missile upon application of 115 volts, 400 Hz, single-phase power defined in 3.4 to supply low-voltage DC and high-voltage DC specified herein (4.4.2.4).

3.4.3.1 Low-voltage supply The low-voltage supply shall operate as specified herein and shall apply over all service conditions and input voltage/frequency combinations.

a. The supply shall be regulated to provide low-voltage DC with the loads (I amps) and DC voltage (Vo) outputs as follows:

LOAD	I(amps)	VoMIN	VoNOM	VoMAX
NO LOAD	0.0	24.0	26.0	27.0
MIN	0.5	24.0	25.0	26.4
FULL	1.5	24.0	25.0	26.4
STRESS 1/	4.0	21.0	25.0	26.4
OVERLOAD 1/	greater than 4.49	0.0	NA	1.0

1/ 30 sec. continuous after power supply turn on, and 10 percent duty cycle (3 sec. on, 27 sec. off) thereafter.

b. The supply shall be less than 1.0 VDC for 0.1 (+0.05, -0.02) seconds after application of input power. The output voltage and current shall then rise in a linear or simple exponential form to its rated voltage and current within 0.2 to 1.0 sec. This shall apply for loads ranging from 0.0 to 4.0 amps.

c. The supply output shall not be less than -1.0 VDC at turn-on, turn-off, all load conditions, and input conditions.

d. The output voltage shall be not greater than 32.0 VDC under any conditions.

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e. The supply shall resume steady state operation within 25 ms after any step change in load current of up to ±2.Ø amps. The maximum voltage output during the transient shall be not greater than 29 VDC and the minimum voltage output not less than 22 VDC.

f. The maximum ripple with a standard steady state input and 1.5 amp load shall be less than Ø.15 volts peak-to-peak. The ripple shall be not greater than 1.Ø volts peak-to-peak for any other condition of input voltage, frequency and load. Ripple frequencies other than fundamental through the tenth harmonic as well as the fundamental frequency of a switching supply, if used, shall be ignored. High frequency noise spikes of switching supplies shall be less than Ø.84 volts peak-to-peak. The ripple voltage shall be not greater than Ø.Ø5 volts rms for loads from Ø.Ø to 1.5 amp and Ø.3 volts rms for loads greater than 1.5 amps in switching supplies.

g. Short circuit requirements are as stated in 3.4.4.

3.4.3.2 High-voltage supply. The high-voltage supply shall operate as specified herein and shall apply over all service conditions and input voltage/frequency combinations. The output shall be at J2 pins 29, 31 and 33.

a. The supply shall be regulated to provide high-voltage DC with the loads (I ma) and DC voltage (Vo) outputs as follows:

LOAD	I(ma)	VoMIN	VoNOM	VoMAX
NO LOAD	Ø.Ø	160.Ø	175.Ø	200.Ø
MIN	125	160.Ø	175.Ø	190.Ø
FULL	175	160.Ø	175.Ø	190.Ø
STRESS 1/	280	160.Ø	175.Ø	190.Ø
OVERLOAD 1/	greater than 320	Ø.Ø	NA	9.Ø

1/ 60 sec. continuous after power supply turn-on and 10 percent duty cycle (3 sec. on, 27 sec. off) thereafter.

b. The output voltage shall be less than the low-voltage supply until the low-voltage supply output is greater than 19.Ø (± 1.Ø) VDC.

c. The output voltage shall be at full rated voltage no later than 2.Ø seconds after application of input power.

d. The output voltage shall be not greater than 260 VDC or shall not be less than -1.Ø VDC under any conditions.

e. The supply output shall be less than 19 VDC whenever the low-voltage supply output is short circuited or the low-voltage supply drops below 19.Ø (± 1.Ø) VDC for longer than 1.Ø second.

f. The supply shall resume steady state operation within 25 ms after any step change in load current of up to 200 ma. The maximum voltage output during the transient shall not be greater than 205 VDC and the minimum voltage output not less than 145 VDC.

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g. The maximum ripple with a steady state input and a 175 ma load shall be less than Ø.6 volt peak-to-peak. The ripple shall be not greater than 1.5 volts peak-to-peak under any condition of input voltage, frequency and load. Ripple frequencies other than fundamental through the tenth harmonic as well as the fundamental frequency of a switching supply, if used, shall be ignored. High frequency noise spikes of switching supplies shall be less than 1.25 volts peak-to-peak under any condition and the ripple shall be not greater than 5ØØ millivolt (mV) peak-to-peak for loads ranging from Ø-175 ma.

h. Short circuit requirements are as stated in 3.4.4. The short circuit shall be at J2 pins 33, 31 and 29.

3.4.4 Short circuit protection. No damage shall occur when the high voltage (175 VDC), low voltage (25 VDC) or both supply outputs are continuously shorted to ground or each other. The primary input current shall be not greater than 33Ø ma when both supplies and the audio amplifier are shorted to ground simultaneously. The outputs shall return to specified values upon removal of all short circuits (See 4.4.2.4).

3.4.5 Firing circuit. The firing circuit shall be constructed to operate (See 4.4.2.5) using electrical and mechanical relays as indicated herein (See Figure 9). Firing circuit loads are as stated below:

Location	Load (±5%)
a. J2 pin 3	27.Ø ohms resistive
b. J2 pin 13	2.5 ohms resistive
c. J2 pin 19	6.2 ohms resistive
d. J2 pin 31	Short to ground
e. J1 pin 33	Ø.5 ohms resistive
f. J1 pin 28	Ø.5 ohms resistive
g. J1 pin 32	5.Ø ohms
h. J2 pin 18	15.Ø ohms resistive
i. J1 pin 3Ø	7.35 ohms resistive

3.4.5.1 Relay K3. Relay K3 (See Figure 9) shall close under either of the following conditions of 3.4.5.1.1 and 3.4.5.1.2.

3.4.5.1.1 Missile regulator load. Relay K3 shall pull in when a true rms voltage of not greater than 4Ø volts at a frequency of 5.7Ø ± Ø.Ø5 kHz is applied between J2 pin 32 and ground. Response shall be limited to fundamental frequency of a sine or square wave missile signal not greater than 4Ø volts rms. Transients generated by the circuit shall be limited to 175 ± 23 volts peak. Relay K3 shall not pull in when a true rms voltage of less than 8Ø volts at 4.Ø ± .Ø5 kHz is applied between J2 pin 32 and ground. Maximum input current shall be 7Ø ma.

3.4.5.1.2 Thermal battery. Relay K3 shall pull in and hold prior to 18 VDC (3.2.5.2) on the application of a ramped voltage at pin 27 of J2. The ramped voltage shall have a slope of 175 (± 1Ø percent) VDC per second with a 21 VDC maximum steady state value. The input impedance of the power supply at pin 27 of J2 shall be 14Ø to 11,ØØØ ohms.

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3.4.6 Feedthrough wiring. Feedthrough wiring resistance shall be not greater than Ø.7 ohms, including shield and other braids and chassis ground (4.4.2.6).

3.4.7 Surge suppression. The output voltages of the low-voltage supply and the high-voltage supply shall be not greater than 32 VDC and 26Ø VDC respectively during surges in the 4ØØ Hz supply voltage as specified in MIL-STD-7Ø4 (4.4.2.7).

3.4.8 Audio amplifier. The audio signal from pin 15 of connector J2 shall be amplified and the output routed to pin 18 of connector J1 (4.4.2.8).

a. These requirements shall apply throughout the power supply operating temperature range as defined in 3.5.1 with a 6ØØ ohm (±1 percent) resistive load.

b. The audio amplifier shall have a voltage gain of 1.Ø (±5 percent) over the frequency range of 4ØØ to 6ØØØ Hz and input range of Ø.5 to 15 volts rms. The gain shall be greater than Ø.25 and less than 1.Ø between the frequency range of 6ØØØ HZ and 2Ø KHZ over the Ø.5 to 15 volts rms input range.

c. Input impedance shall be 2ØK ohms (±1 percent).

d. Output impedance shall be less than 25 ohms.

e. Output distortion shall be not greater than 1Ø percent for any input from Ø.5 to 15 volts rms over a frequency range of 8ØØ to 6ØØØ Hz.

f. With the input shunted by a 2ØK ohm (± 1 percent) resistor, the random noise across the load shall be not greater than Ø.15 volts rms.

g. A short circuit or open circuit across the audio amplifier output shall not damage the amplifier. The amplifier shall operate as specified after the short circuit or open circuit is removed.

h. The maximum output of the audio amplifier shall be limited to 6Ø volts peak to peak with any input voltage from Ø.Ø to 26 volts rms.

i. The output of the audio amplifier shall be phase shifted 18Øo ± 2Øo from the input to the audio amplifier over the frequency range of 15ØØ Hz to 6ØØØ Hz.

3.4.9 Accrued test time. No power supply to be furnished shall have accrued more than 1ØØ hours unless approved by the procuring activity.

3.4.1Ø Fit. The power supply shall satisfy the fit test paragraph 7-1117 of NAVAIR 11-75A-54 using the acceptance gauge illustrated in Figure 3-13 of NAVAIR 11-75A-54.

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3.5 Service conditions. The power supply shall be so designed and constructed that no fixed part or assembly shall become loose, no moving or movable part, or control be shifted in setting, position, or adjustment, and no degradation be caused in the performance beyond that specified herein when subjected to the service environment specified in MIL-E-5400, for Class 2 equipment (except as specified herein). Any combination of the conditions specified herein may apply.

3.5.1 Temperature. The power supply shall operate when exposed to the temperature conditions specified in MIL-E-5400, for Class 2 equipment except the upper temperature limit shall be +85° C and the lower limit shall be -54° C. The temperature within the specified ranges may remain constant for long periods of time and may vary at a rate as high as 1 degree per second.

3.5.2 Altitude. The power supply shall operate when exposed to the altitude conditions specified herein. The altitude may remain constant for long periods and vary at a rate as high as 0.5 inch of mercury per second.

Altitude extremes: sea level (30.0 in Hg) to 70,000 feet (1.32 in Hg).

3.5.3 Temperature-altitude combination. The power supply shall operate under the temperature-altitude combinations shown on FIGURE 1.

3.5.4 Humidity. The power supply shall withstand the effects of humidities up to 100 percent, including conditions wherein condensation takes place in and on the power supply. The equipment shall withstand these conditions both during operation and nonoperation.

3.5.5 Vibration. The power supply shall be designed and constructed so that it will suffer no damage or fail to meet performance specifications when subjected to the following random and sinusoidal vibration environments:

a. Random excitation. A random vibration spectrum extending from 20 to 2000 Hz at 0.072 to 0.233 G^2/Hz.

b. Sinusoidal excitation. Sinusoidal vibration covering the frequency range from 5 to 500 Hz at input levels up to $\pm$ 10g.

3.5.6 Service shock. The power supply shall not suffer damage or fail to provide required performance after sustaining shocks up to and including 20g along any axis.

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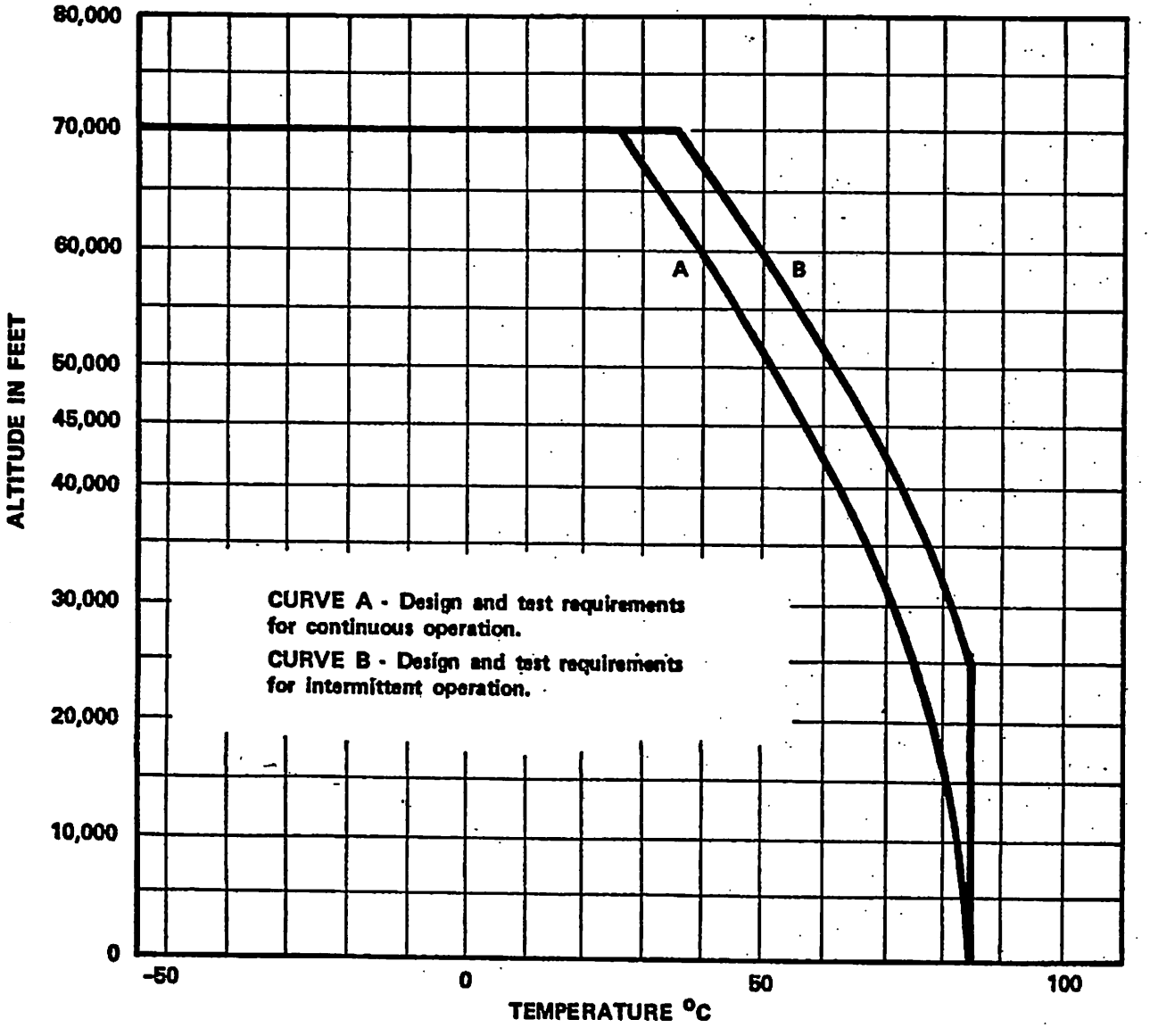


FIGURE 1. Operational requirements for temperature vs. altitude.

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3.5.7 Salt fog. The power supply shall withstand exposure to a salt-sea atmosphere consisting of a 5±1% solution for up to 168 hours (See 4.4.3.5).

3.5.8 Acceleration. The power supply shall meet all requirements of this specification when subjected to accelerations up to 12 g's for periods greater than 1 minute (See 4.4.3.7).

3.6 Electromagnetic interference. The contractor shall prepare an Electromagnetic (EMI) Control Plan (see 6.2.2) using MIL-HDBK-241 as a design guide. The power supply shall meet the requirements of MIL-STD-461 for class A1b equipment, tests CE03, CE07, CS02, CS06, RE02, RS02 and RS03 apply (See 4.4.3.6).

3.7 Reliability. Reliability requirements are as specified in 3.7.1 through 3.7.3.

3.7.1 Quantitative reliability requirements. The Mean Time Between Failure (MTBF) for the power supply, when operated under any combination of the loads and environmental conditions specified herein, shall have a lower test value of 1000 hours (θ_1) and an upper test value of 2000 hours (θ_0) (4.4.4 and 4.5.4).

3.7.2 Reliability plans and reports. The manufacturer shall prepare reliability plans and reports in accordance with tasks 101 through 105, 203, 204, 206 through 209, and 301 through 304 of MIL-STD-785 (see 6.2.2).

3.7.3 Long-term storage. The probability that the power supply survives one year in storage without degradation shall be no less than 0.9992 as determined by applying Handbook LC-78-1.

3.8 Maintainability. The power supply shall require no external adjustment or scheduled maintenance.

3.9 Drawings. Engineering drawings of the power supply shall be prepared in accordance with the requirements of DOD-STD-100 and DOD-D-1000, Level 2 (6.2.2).

3.10 Government-loaned property. Government loaned property will be as follows:

ITEM	PART NUMBER
LAU-7 series Missile Launcher	58A164H891
LAU-7 Nitrogen Receiver	58A164D556

3.11 JAN marking. The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of military specifications. Accordingly, items procured to, and meeting all of the criteria specified herein and in applicable detail specifications shall bear the certification mark "JAN" except that items too small to bear the certification mark "JAN" shall bear the

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letter "J". The "JAN" or "J" shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or "J" may be located on the first line above or below the part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in applicable detail specifications shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets or detail specifications, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products procured to contractor drawings or specifications. The United States Government has obtained Certificate of Registration No. 504,860 for the certification mark "JAN".

3.12 Workmanship. The power supply including all parts and accessories shall be constructed and finished in a manner that shall assure compliance with all requirements specified herein. Unless otherwise specified herein, fabricating and assembly practices shall be in accordance with MIL-STD-454 requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, and shall be approved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure conformance to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in 4.4 shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor shall be subject to procuring activity verification at any time. Verification shall consist of the functions specified in 4.1.2.1 and 4.1.2.2.

4.1.2.1 Surveillance of operations. Operations shall be closely observed or monitored to determine that practices, methods, and procedures as specified herein are being implemented and adhered to.

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4.1.2.2 Product inspection. Inspections shall be performed to ensure the quality of the power supply to be offered for acceptance. Failure of the contractor to promptly correct deficiencies either discovered by him, or of which he is notified, shall be cause for suspension of acceptance until corrective action has been made or until conformance of product with prescribed criteria has been demonstrated.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed under the standard conditions specified in 3.3.2.4.

4.4 Qualification inspection. Qualification inspection shall be performed on a qualification power supply sample produced with equipment and procedures normally used in production.

4.4.1 Qualification sample. Prior to delivering the qualification power supply sample to a test activity designated by the qualifying activity (see 6.3), the manufacturer shall have conducted a Reliability Development Test (RDT) in accordance with 4.4.6. The qualification sample, consisting of 12 power supplies which have been subjected to the tests specified in 4.5.2 and 4.5.3, shall be delivered to the test activity to be inspected and tested as specified in 4.4.1.1.

4.4.1.1 Qualification inspection routine. The qualification sample will be subjected to the inspections specified in Table III. The number of power supplies for each test are as stated in Table III. The tests of 4.4.3 shall be completed prior to the start of test 4.4.5.

4.4.1.2 Power supply failure. In the event of a failure (6.5) during qualification inspection, the following action shall be taken:

a. The test activity will halt testing and immediately notify the qualifying activity.

b. The failed power supply will be returned to the manufacturer for failure analysis. Upon receipt of a failed power supply, the manufacturer shall take the following actions:

(1) Invite a qualifying activity representative to be present during all portions of failure analysis.

(2) Determine the cause and analyze the physics of the failure

(3) Perform a failure analysis.

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(4) Prepare failure report (see 6.2.2).

c. The qualifying activity will determine whether testing is to continue.

d. Upon completion of qualification inspection, the test activity shall prepare a test report, including EMI test report.

4.4.1.3 Qualification rejection. If at any time during or after qualification inspection the qualifying activity determines a failure to be a recurring manufacturing problem, design deficiency, flight safety related problem, or repetitive failure. Qualification shall be halted and qualification shall not be granted. A power supply removed from testing by the qualifying activity during the tests of 4.4.2 or 4.4.3 shall be cause for refusal to grant qualification. Power supplies removed from Reliability Qualification Test (See 4.4.5) in excess of allowed test failures shall also be cause for refusal to grant qualification. At the completion of the tests of 4.4.2 and 4.4.3, the manufacturer shall prepare failure analysis reports (see 4.4.1.2(b)) on each power supply failure.

4.4.1.4 Retention of qualification. To retain qualification, the contractor shall forward a report at 22-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of a summary of any tests performed (See 4.5.3 and 4.5.4) for delivery over the period, indicating at a minimum the number of power supplies that have passed, the number that have failed, the specific tests each power supply failed, the part(s) failed and the cause of failure. The results of tests of all reworked power supplies (See 4.5.4) shall be identified and accounted for. The summary shall include corrective actions. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified product list. Failure to submit the report within 30 days after the end of each 22-month period may result in loss of qualification for the power supply. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 22-month period that the inspection data indicates failure of the qualified power supply to meet the requirements of this specification. In the event that no production has occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the power supply. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a qualification power supply sample for testing in accordance with the qualification inspection requirements.

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TABLE III. Qualification inspection.

No. of units	Test	Requirement Para	Test Para.
All	Individual equipment tests	3.4	4.4.2
4	Specification conformance tests	3.5	4.4.3
	Temperature-altitude combinations	3.5.1	4.4.3.1
		3.5.2	
		3.5.3	
	Humidity	3.5.4	4.4.3.2
	Vibration	3.5.5	4.4.3.3
	Service shock	3.5.6	4.4.3.4
	Salt-fog	3.5.7	4.4.3.5
	Electromagnetic interference	3.6	4.4.3.6
	Acceleration	3.5.8	4.4.3.7
10	Reliability qualification	3.8	4.4.5

4.4.2 Individual equipment tests. Each power supply shall be subjected to the individual equipment tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each power supply shall pass the following tests:

	Requirement Para.	Test para.
a. Examination of product	3.14	4.4.2.1
b. Insulation resistance test	3.4.1	4.4.2.2
c. Gas pressure system test	3.4.2	4.4.2.3
d. Operation test	3.4.3, 3.4.4	4.4.2.4
e. Relay operation test	3.4.5	4.4.2.5
f. Feedthrough wiring test	3.4.6	4.4.2.6
g. Surge test	3.4.7	4.4.2.7
h. Audio amplifier test	3.4.8	4.4.2.8
i. Meter inspection	3.4.9	4.4.2.9
j. Fit test	3.4.10	4.4.2.10

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4.4.2.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.2.2 Insulation resistance test. An insulation resistance test shall be performed in accordance with MIL-STD-202, Method 302. A high potential is applied between each isolated circuit and all other isolated circuits, including ground. Insulation resistance shall be not less than 100 megohms at 500 (plus 50 or minus 20) VDC. Circuits which will be damaged by this test shall not be subjected to this test.

4.4.2.3 Gas system test. The gas system shall be tested with dry air or nitrogen at a pressure of 3800 (± 50) psig and 2200 (± 50) psig at 77 (± 10)o fahrenheit for leakage and at a pressure of 3000 (± 100) pounds for flow.

a. With a suitable pressure system at a pressure of 3800 (± 50) psig and 2200 (± 50) psig, a leakage test shall be performed and the maximum leakage rate shall be not greater than 1 cubic centimeter per minute at each pressure.

b. A pressure drop shall be performed with a suitable rotometer and pressure monitoring system. With the inlet of J1 maintained at a pressure of 3000 (± 100) psig at a flow rate of 28.3 standard liters per minute, the total pressure drop across the feedthrough system shall be less than 50 psig.

c. The pressure relief valve shall be checked to verify that the valve shall operate as prescribed (See 3.4.2).

4.4.2.4 Operation tests. The power supply shall be tested to verify operation of the low-voltage supply and high-voltage regulated supplies. Unless otherwise stated the input conditions for each test shall be as follows:

Voltage (VAC)	Frequency (Hz)
98	380, 400, 420
115 (standard)	380, 400 (standard), 420
124	380, 400, 420

a. For each load condition verify the requirements of 3.4.3.1a and d and 3.4.3.2.a and d.

b. For full and stress load conditions verify the requirements of 3.4.3.1b and 3.4.3.2b and c.

c. Apply a full load to the low-voltage supply. Switch the load to 0.5 amp within 0.2 ms. Verify the requirement of 3.4.3.1.d and e. Switch the load back to 1.5 amp within 0.2 ms and reverify the requirements.

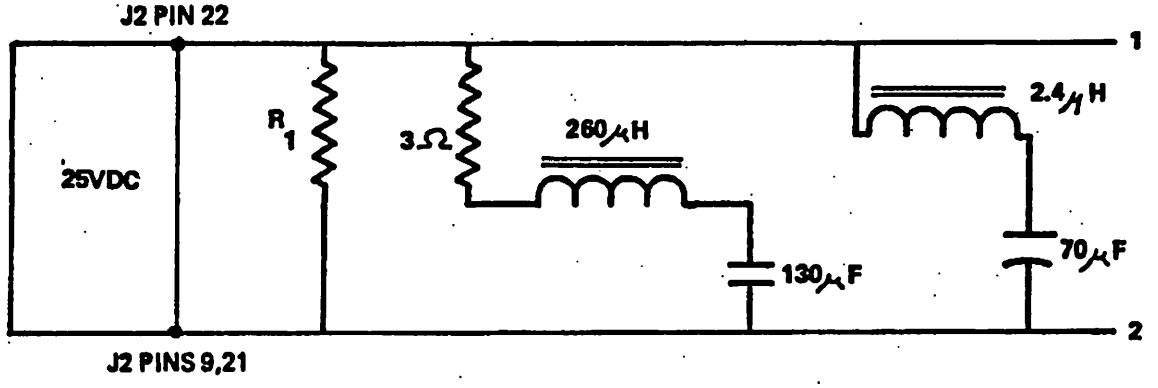
d. Apply the full load to the high-voltage supply. Switch the load to 125 ma within 0.2 ms. Verify the requirements of 3.4.3.2d and f. Switch the load back to 175 ma within 0.2 ms and reverify the requirements.

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e. With the load on the low-voltage supply as shown on FIGURE 2, verify the requirements of 3.4.3.1b, c, d and f with R1=18 ohms (1.5 amp) and 6 ohms (4 amp). The 4 amp load shall have a 10 percent duty cycle. Measurements shall be taken at points 1 and 2.

FIGURE 2. 25 VDC Load.

NOTE: tolerance shall be ± 2%



NOTE: Tolerance shall be ± 2%

FIGURE 2. 25VDC load.

f. With loads of 175 ma and 280 ma (10 percent duty cycle) verify the requirements of 3.4.3.2g.

g. With standard inputs, short the low-voltage supply to ground for 60 seconds. Verify 3.4.3.2e. Adjust the input voltage, reducing the low-voltage supply output level to verify 3.4.3.2e.

h. Short each supply to ground, short the low-voltage supply to the high-voltage supply and short both supplies to ground. Each test shall last 60 seconds. Verify 3.4.4.

4.4.2.5 Relay operation. With 115 VAC, 400 HZ applied to J1 pin 7, and utilizing the high-voltage and low voltage loads of 3.3.2.3(f), the relay firing circuit shall be tested as specified herein.

a. With inputs as listed in TABLE IV and "A" equal to 18 (+5, -0.0) VDC, verify the outputs in TABLE IV.

b. Apply loads as specified in 3.4.5. Apply, with a 10 percent duty cycle and "power on" time to be 0.9 ± 0.1 seconds, the input conditions of TABLE IV with the "A" equal to 18 (+0.5, -0.0) VDC verify the outputs of TABLE IV with the "A" conditions being the voltages specified in 3.4.5 at the 18 VDC test.

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c. Repeat test b. above except use 29.5 (+Ø.5) VDC for the "A" input condition, and verify the outputs with the "A" condition being specified in 3.4.5 at the 29.5 VDC test.

TABLE IV. Relay performance.

A = 18 to 3Ø VDC C = para. 3.4.5.1.1
B = 175 (±5) VDC D = less than Ø.ØØØ1 amps DC when shorted to ground
E = missile regulator load para. 3.4.5.1.2
NOTE: all outputs shall be ± 7%

J1 inputs							J2 in		J1 outputs					J2 outputs						
2	5	6	11	15	21	25	27	32	28	29	30	32	33	3	13	18	19	23	30	31
	A								A	D	A	D	A	D	D	D	D	D	D	B
			A	A					D	A	A	D	D	D	D	A	D	D	D	B
			A	A	A				D	A	A	D	D	D	D	A	D	A	A	B
		A	A	A	A				D	A	A	D	D	D	D	A	D	D	D	B
			A	A	A	A			D	A	A	D	D	D	D	A	D	A	D	B
A			A	A	A	A			A	A	A	D	D	A	A	A	D	A	D	D
A			A	A	A	A	C		A	A	A	A	A	A	A	D	A	A	D	D
				A					D	A	D	D	D	D	D	A	D	D	D	B
A			A	A	A	A		E	A	A	A	A	A	A	A	D	A	A	D	D

4.4.2.6 Feedthrough wiring. All feedthrough wiring and chassis ground shall be tested for continuity. Resistance shall be not greater than Ø.7 ohm, including shield and other braids.

4.4.2.7 Surge test. The surge test shall consist of monitoring the peak output voltages of the 25.2 and 175 VDC circuits during changes in the 4ØØ Hz supply voltage. A total of six cycles shall be required. The test shall be conducted with no load on the low-voltage (25.2 VDC) and high-voltage regulated (175 VDC) circuits. The 4ØØ cycle input voltage is raised from Ø to 19Ø volts rms, held for 1ØØ ± 5 milliseconds, and dropped to 122 volts rms. Three cycles are required. The second three cycles consist of raising the input voltage from 122 volts rms to 19Ø volts rms, holding for 1ØØ ± 5 ms, and dropping to 122 volts rms. The rise and fall times of the input voltage shall be not greater than 7.5 milliseconds. The requirements of 3.4.7 shall be met. After the test the power supply shall meet the requirements of 4.4.2.4.

4.4.2.8 Audio amplifier test. The audio amplifier shall be tested as specified herein.

a. With 115 volts, 4ØØ Hz input applied to pin 7 of J1 and a 6ØØ (1 percent) ohm load at pin 18 of J1, the audio amplifier shall meet the requirements of 3.4.8 (b, c, d, e and i) when tested with an input at pin 15 of J2 as follows:

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TEST NO.	INPUT VOLTAGE (V rms ±5%)	FREQUENCY (Hz ±1%)
1	Ø.5	4ØØ
2	Ø.5	15ØØ
3	Ø.5	6ØØØ
4	5	15ØØ
5	1Ø	4ØØ
6	1Ø	15ØØ
7	1Ø	6ØØØ
8	15	15ØØ
9	15	4ØØ
1Ø	15	6ØØØ
11	1Ø	2Ø,ØØØ

b. With the input shunted by a 2ØK ohm resistor, the requirements of 3.4.8f shall be met.

c. A short circuit shall be applied to the audio amplifier output with the input of 15 Vrms at 15ØØ Hz for 6Ø seconds. Upon removal of the short circuit the audio amplifier shall operate as specified in 3.4.8.

d. With a 25 volt rms input at 4ØØ, 15ØØ and 6ØØØ Hz verify the audio output is less than 5Ø volts peak to peak.

e. Repeat c above with no load applied for 6Ø seconds.

4.4.2.9 Fit test. The power supply shall be mounted in the minimum enveloped-sized jig (see 3.4.1Ø). Light shall be visible around all sides of the power supply except the bottom.

4.4.3 Specification conformance tests. Four power supplies shall be subjected to all environmental test specified herein except two power supplies completing humidity (See 4.4.3.2) shall not complete salt fog (See 4.4.3.5). Vibration (See 4.4.3.3) and shock (See 4.4.3.4) shall be conducted prior to other tests.

4.4.3.1 Temperature-Altitude. Tests for temperature and temperature-altitude requirements (See 3.5.1, 3.5.2, and 3.5.3) shall be as specified in TABLE IV. The rates of change of temperature and pressure shall be the maximum attainable by the equipment but shall not be less than 5º C per second. The power supply shall be operating throughout the test except as noted.

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TABLE V. Temperature - altitude tests.

	STEP	TEMP (°C)	ALTITUDE (FT)	TIME (HR)	NOTE
START	1	ROOM	SITE		1/
	2	-54	SITE	2.0	
	3	-54	70,000	0.5	
	4	-10	SITE	1.0	
	5	ROOM	SITE	0.5	
	6	85	SITE	16.0	
	7	71	SITE	4.0	
	8	85	SITE	0.5	
	9	36	50,000	4.0	
	10	60	50,000	2.0	2/
	11	10	70,000	4.0	
	12	35	70,000	2.0	2/
END	13	ROOM	SITE		

1/ The power supply shall be de-energized during all chamber temperature adjustments.

2/ The power supply shall be energized for four 15 minute periods separated by four 15 minute de-energized periods.

4.4.3.2 Humidity. Tests for conformance with humidity operating specifications (See 3.5.4) shall be conducted in accordance with MIL-STD-810, method 507.2, procedure III. This power supply shall not complete Salt-fog (See 4.4.3.5). 10 cycles shall be performed.

4.4.3.3 Vibration. The power supply shall be operating during vibration tests. The output at J1 pin 33 shall be continuously monitored for inadvertant missile launch. The power supply shall be attached to the exciter by means of a rigid fixture. The vibration input shall be measured on the fixture adjacent to the power supply center.

a. Random vibration. Random vibration shall consist of 0.5 hour at endurance levels, 1.0 hour at functional levels followed by 0.5 hour at endurance levels. The test spectrum is shown on FIGURE 3 and shall be performed along each orthoganal axis.

	W1	W2	f1	f2	Level
Endurance	0.072	0.233	500	1500	20g rms
Functional	0.020	0.065	500	1500	10g rms

b. Sinusoidal vibration. Sinusoidal vibration shall be performed as specified in MIL-L-22620 except that the power supply shall not be mounted in the launcher and the frequency shall be not greater than 500 Hz.

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4.4.3.4 Shock. Tests for conformance with service shock requirements (See 3.5.6) shall be conducted as specified herein. The power supply shall be subjected to the required accelerations in both directions along each of the three axes. The peak applied values shall be within plus 15, minus Ø percent of the required accelerations.

a. Functional Shock. The shock pulse shall be a half-sine pulse. The spectrum from FIGURE 4 shall be employed as the test spectrum for each axis. The effective (above one-third peak value) duration of the test shock time history shall fall between six and nine ms. Peak acceleration shall be 2Ø g's. Three shocks shall be performed for each of the three axes. Wave input shall be a damped sine wave with a quality factor (Q) of 2Ø. The pulse shall reach peak value in not more than Ø.75 sec. or less the Ø.4 sec.

b. Catapult and arrested landing. The power supply shall be shocked in both directions along the longitudinal axis only. The peak value shall be ±5 g's. The acceleration shall reach peak value in not more than Ø.75 seconds or less than Ø.4 seconds. Duration shall be 2.Ø seconds. The shock shall be applied 3Ø times with a 1Ø second rest time between each shock. The shock input may be either a half-sine pulse or a full-sine pulse.

4.4.3.5 Salt-fog. Tests for conformance with salt-fog requirements (See 3.5.7) shall be in accordance with MIL-STD-81Ø, method 5Ø9.2, procedure I. Dust caps and protective covers shall be in place. The power supply shall not be operated during this test.

4.4.3.6 Electromagnetic-interference. Tests for conformance with electromagnetic-interference (EMI) requirements (See 3.6) shall be in accordance with MIL-STD-462. The power supply shall be mounted in a LAU-7 launcher for EMI testing. Data shall be obtained using the full load conditions of 3.3.2.4. Susceptibility shall be measured at the standard missile umbilical of the launcher.

4.4.3.7 Acceleration. The power supply shall be tested in accordance with MIL-STD-81Ø, method 513.3, procedure II. The test shall be 1.Ø (+Ø.25) minute duration for each axis. The fore and aft acceleration shall be 1Ø (±1) g's and the up/down and lateral accelerations shall be 12 (±1) g's. The fore and aft axis is defined as the long axis.

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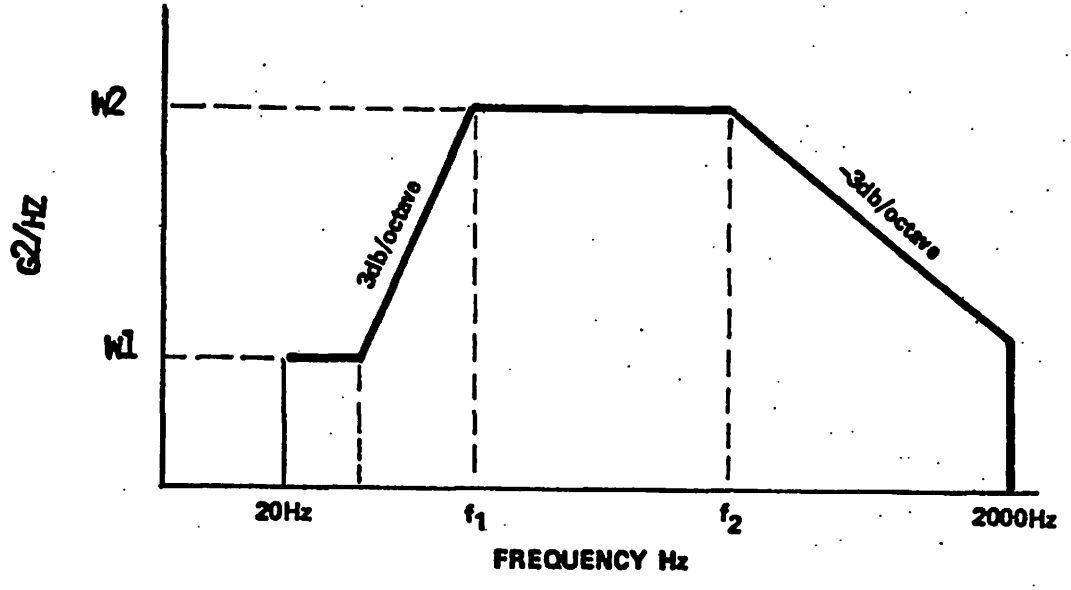


FIGURE 3. Random vibration spectrum.

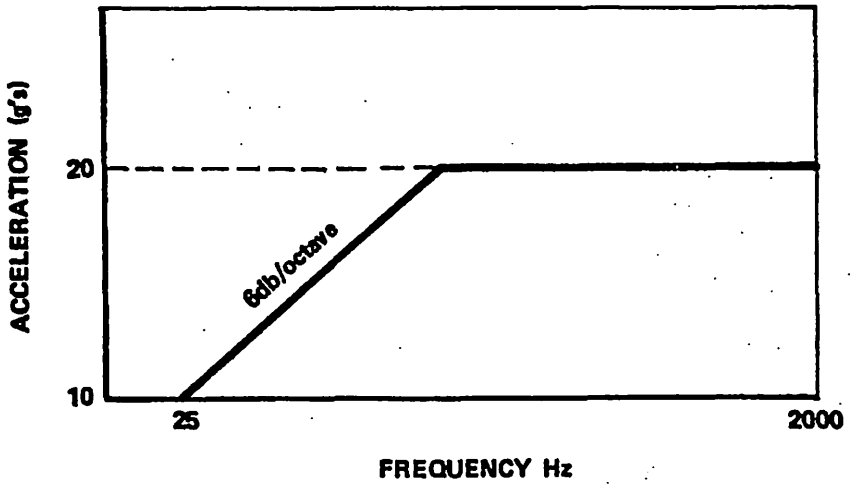


FIGURE 4. Test shock response spectrum

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4.4.4 Reliability qualification test. A reliability qualification test shall be performed on not less than 10 qualification power supply samples in accordance with MIL-STD-781, Test Plan XIV C. The upper test MTBF shall be 2000 hours and the lower test MTBF shall be 1000 hours (3.7.1).

4.4.4.1 Test cycle. A test cycle shall be as shown on FIGURES 3, 6 and 7. Three cycles of altitude testing as shown on FIGURE 6 shall be performed prior to the temperature and vibration cycling of FIGURES 3 and 7. The vibration spectrum shall conform to the functional test of 4.4.3.3a.

4.4.4.2 Performance monitoring. The 25.2 and 175 VDC supply outputs, relays and the audio amplifier output shall be monitored continuously for compliance with specified requirements. In addition, a complete performance test of all functional requirements as specified in 4.4.2.4, 4.4.2.5, 4.4.2.7, and 4.4.2.8. shall be performed at least daily on each unit at the hot and cold temperature extremes.

4.4.4.3 Inputs and loads. During the test the input and output loads shall be as specified in 3.3.2.3.

4.4.5 Reliability development test (RDT). The manufacturer shall conduct a 10,000-hour RDT in accordance with MIL-STD-2068 to be witnessed by the qualifying activity or its designate. A minimum of four and a maximum of ten power supplies shall be tested. The test power supplies shall be representative of production power supplies and shall have completed the individual equipment tests specified in 4.4.2. The purpose of this test is to mature the design of the power supply by exposing it to simulated environments including temperature cycling and random vibration, simultaneously. The environmental profile for this test shall be those of FIGURES 3 and 7. Power supply loads are as stated in 3.3.2.3. All failures during this test shall be analyzed to determine the causes. Corrective actions to prevent the recurrence of these failures shall be implemented in all RDT units. The corrective actions shall then be tested under the same environmental conditions to assess the effectiveness of the corrective actions. There is no pass-fail criteria for the test; however, the MTBF shall be monitored using the growth model outlined in MIL-STD-2068. At the completion of RDT, the contractor shall prepare an RDT report which includes all failure analyses and corrective actions taken (see 6.2.2). Perform the tests as specified in 4.4.2.5b every 10 minutes during power-on periods. The vibration spectrum of FIGURE 3 shall conform to the functional test of 4.4.3.3a. Only power supply "power on" time shall be counted toward the 10,000 hours. The 400 Hz input shall vary between 98, 115 and 124 V AC with each input maintained for one-third the operating time. Every tenth cycle the maximum temperature shall be +85o C.

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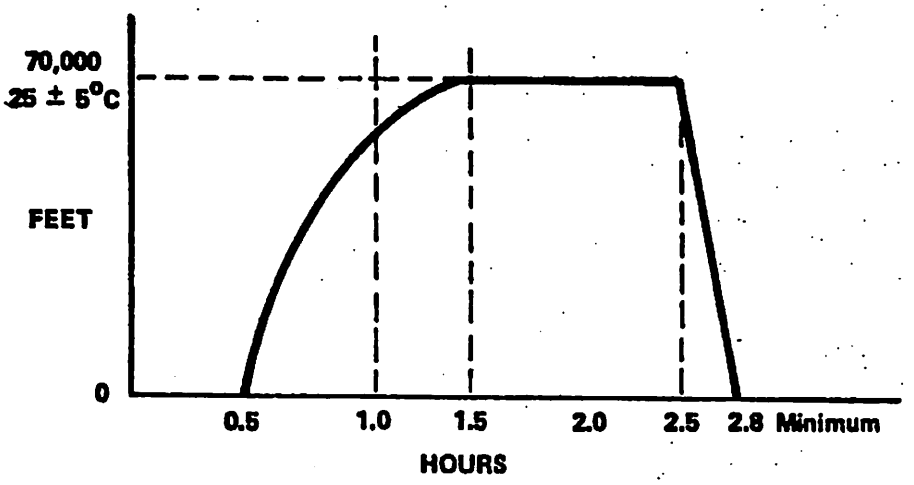


FIGURE 6. Altitude cycle.

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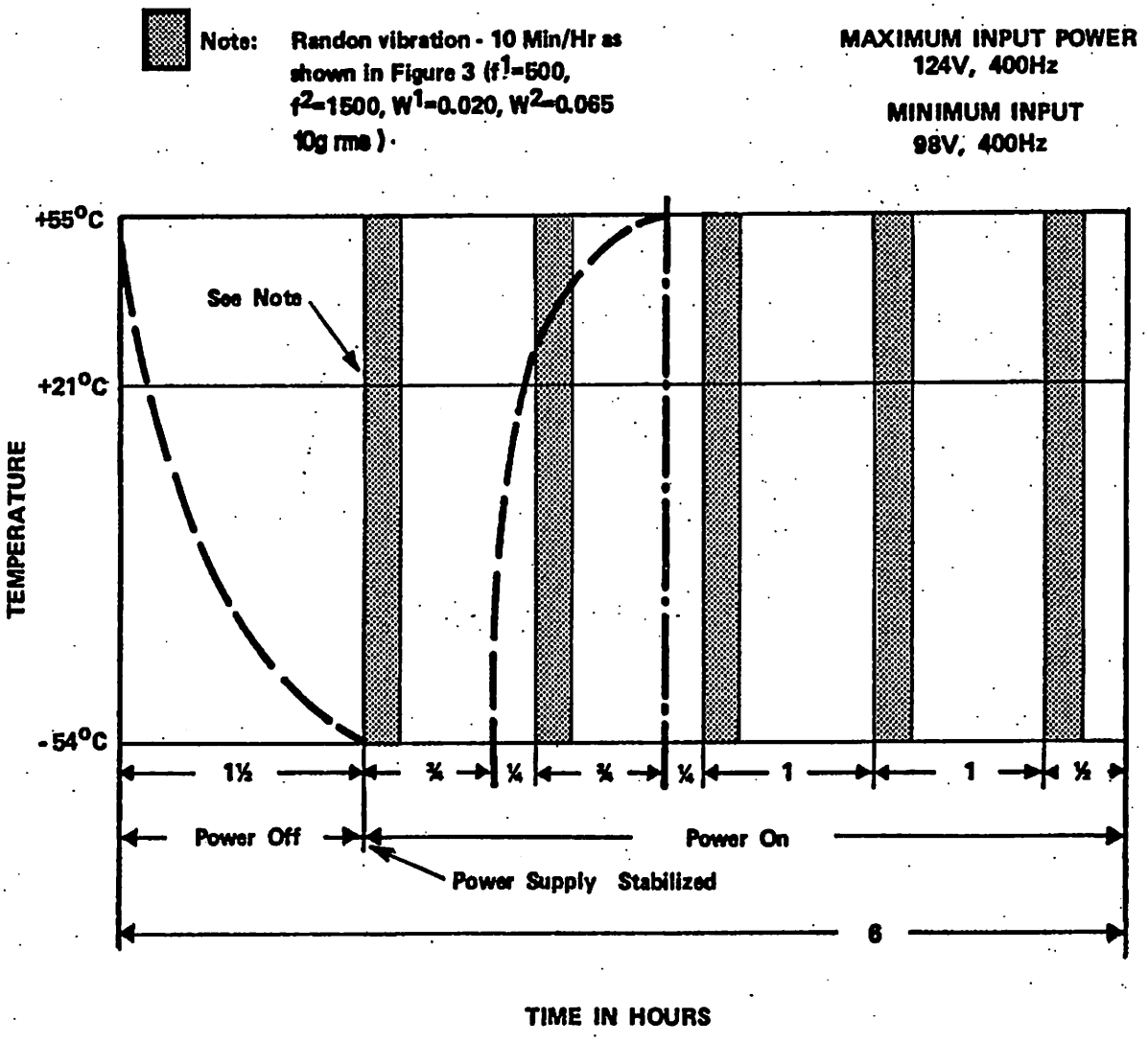


FIGURE 7. Test cycle

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4.5 Quality conformance inspection. Quality conformance testing shall be conducted by the contractor to provide the basis for Government acceptance of production power supplies. The contractor shall prepare test procedures (see 6.2.2) and furnish all test equipment used. All inspection and testing may be subject to monitoring and witnessing by Government representatives. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of acceptance of the finished product. Quality conformance testing shall include all tests deemed necessary by the procuring activity to determine that production power supplies are equivalent to the approved qualification sample in material, workmanship, performance, and reliability, to disclose and correct deficiencies, and to ascertain any impact of incorporated changes. Quality conformance testing shall be comprised of:

a. Component testing (see 4.5.2)
b. Individual testing (see 4.5.3)
c. Packaging inspection (see 4.8)

4.5.1 Quality conformance inspection test data. Reports of individual tests (4.5.3) shall be prepared by the contractor and made available to the government inspector upon request.

4.5.2 Component testing. The following items covered by this specification shall be subjected to the tests specified herein to determine compliance with all applicable requirements. The standard conditions of temperature, humidity, vibration, and altitude shall apply unless otherwise specified. 100 percent screening is required. Lot sampling is not acceptable and shall not be performed by the device manufacturer.

a. Transformers and inductors (see 4.5.2.1)
b. Capacitors and resistors (see 4.5.2.2)
c. Relays (see 4.5.2.3)
d. Transistors and diodes (see 4.5.2.4)
e. Microelectronic Modular assemblies (see 4.5.2.5)

4.5.2.1 Transformers and inductors. Each transformer and inductor shall be inspected to verify conformance of applicable electrical and mechanical requirements. Each item shall be checked to verify that breakdown does not occur when a 500 V rms test voltage is applied between windings or windings and core. For nonstandard magnetics, a temperature cycling test consisting of at least ten cycles between -55° C and 125° C shall be performed.

4.5.2.2 Capacitors and resistors. Tests shall be performed on each capacitor and resistor to verify that values are within tolerance given by manufacturer's specifications. A surge-current test shall be performed on all solid tantalum capacitors. For wirewound resistors and resistors packaged in single inline packages, or dual inline packages, a 70-hour power conditioning test at 70° C shall be performed. For carbon composition resistors, a 100-hour power conditioning test at 100° C shall be performed.

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4.5.2.3 Relay test. Each relay used in the power supply shall be operated to verify pick-up and dropout voltages as specified by the relay manufacturer's specifications after each relay has been stabilized at temperatures of -54º C, +25º C, and +85º C.

4.5.2.4 Transistor and diode test. Each transistor and diode shall be tested in accordance with MIL-STD-750 to verify operating characteristics as published by the manufacturer of the device. For JANTX discrete semiconductor devices, sample electrical and destructive physical analysis (DPA) testing shall be performed on all incoming lots. DPA testing shall include hermeticity, X-ray, de-lid/depot, pull tests on bonds, plating adhesion, solderability, and (as necessary) scanning electron microscope examination. For nonstandard discrete semi-conductors, a minimum of ten temperature cycles and sample DPA including scanning electron microscope examination shall be performed on all incoming lots. Temperature cycling shall be applied on a 100% basis if the part uses gold bonds; aluminum wire bonds with wire diameter less than 2 mil; an electrically-isolated package; external crimp-terminals; clip-contracts; or soft-solder die bonds. A cycle is defined as follows:

The case temperature shall be changed from room ambient to -54º C, from -54º C to +85º C, and from +85º C to room ambient. The minimum rate of change of case temperature shall be 10º C per minute.

4.5.2.5 Microelectronic modular assemblies. Microelectronic modular assemblies shall be tested to the requirements of MIL-STD-883 level B as a minimum. In addition to Class B requirements, all electronic cavity devices (unless passivated) shall be subjected to a particle impact noise detection (PIND) test in accordance with Method 2020, condition A of MIL-STD-883. Hybrid circuits shall be screened to MIL-STD-883, Method 5008. For integrated circuits (IC), 100 percent electrical testing while being exposed to temperature of -55º C, 125º C, and 25º C, and sample DPA shall be performed on all incoming lots. Each IC shall be subjected to 100 temperature cycles before being subjected to electrical testing. For hybrids, 100 percent electrical testing while exposed to temperatures of -55º C, 125º C, and 25º C, 100 percent PIND testing, and sample DPA shall be performed on all incoming lots. Each hybrid shall be subjected to 100 temperature cycles prior to being subjected to electrical testing.

4.5.3 Individual testing. Each production power supply shall be subjected to individual testing. Each power supply shall complete the tests as specified in 4.4.2.1 through 4.4.2.9 and 4.5.4. Unless otherwise specified the standard conditions of 3.3.2.3 shall apply.

4.5.4 Production reliabilitiy acceptance test. Each power supply shall be tested in accordance with the all equipment reliability test of MIL-STD-781 Test Plan XVIIIC. The lower test MTBF shall be 1,000 hours. The test cycle shall be in accordance with Figures 3 and 8. Every third cycle shall have a high temperature of +85ºC vice +55ºC. Performance monitoring shall be in accordance with 4.4.4.2. Each unit shall be tested for at least 10 cycles (45 operating hours) with the last two cycles failure free. All failures shall be evaluated and corrective action determined and implemented as specified in

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4.6. Continuous monitoring of all outputs is required during the last two cycles. In the event of a failure during the last two cycles, the last two cycles shall be repeated until failure free. STRESS loads of 3.4.3.1a and 3.4.3.2a shall be applied at all power supply starts with FULL loads applied thereafter. The tests specified in 4.4.2.5b shall be conducted at 10-minute intervals. All failures shall be evaluated and corrective action determined and implemented as specified in 4.6.

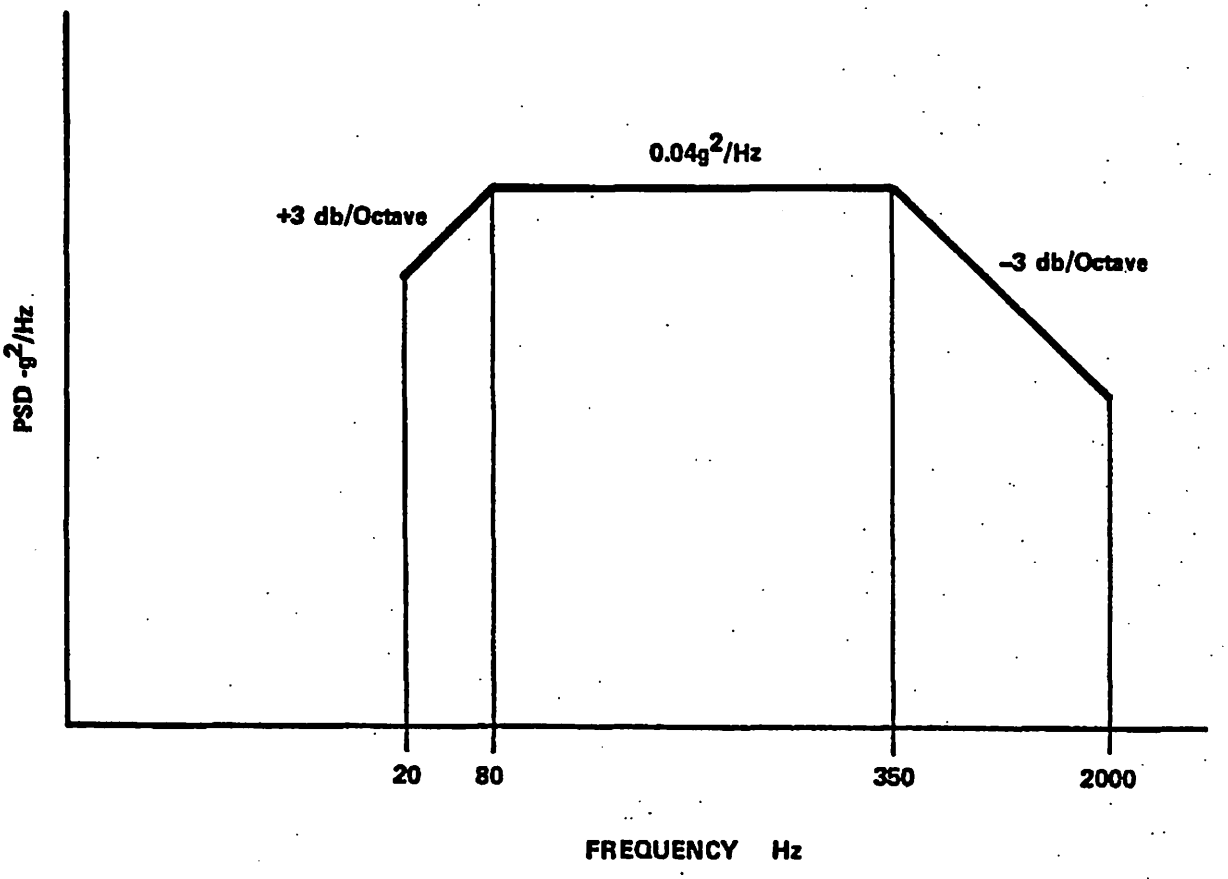
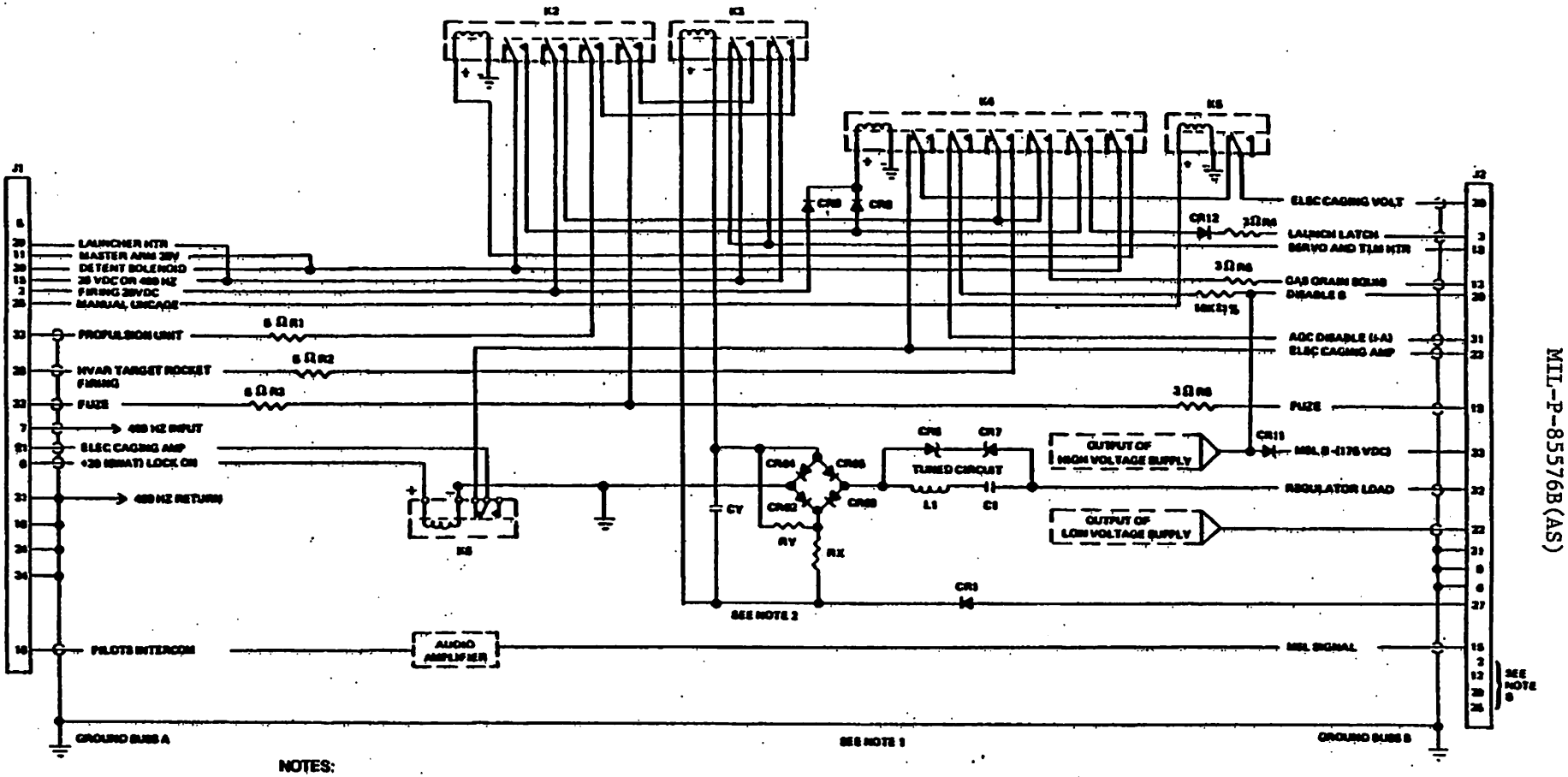


FIGURE 8. Random vibration spectrum (6g rms)
(Production Reliability Acceptance Test)

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NOTES:

1. GROUNDING RINGS AT J1 AND J2 SHALL BE CONNECTED TOGETHER WITH NUMBER 16 AWG STRANDED WIRE AND ONE RING (BUSS A OR B) SHALL BE BONDED TO THE CHASSIS BOTTOM AT THE SELF LOCKING NUT (ITEM 3 OF 60A89D700).
2. RX, RY, CV, CR6, CR7 AND L1 AND C1 SHALL BE SELECTED FOR CONFORMANCE TO THE REQUIREMENTS OF 3.4.5.1.
3. DIODE CR12 SHALL BE IN1614 OR HIGHER CURRENT AND VOLTAGE RATINGS.
4. DIODES CR1 THROUGH CR5, CR8, CR9 AND CR11 SHALL HAVE NOT LESS THAN 1 AMP, 1000 VOLT RATING.
5. ALL REFERENCE DESIGNATORS ARE FOR INFORMATION ONLY AND MAY BE CHANGED DURING DESIGN
6. ALL RELAY COILS SHALL HAVE SURGE SUPPRESSION TO LIMIT VOLTAGES TO ±42 VOLTS PEAK.
7. RESISTORS R1 THROUGH R6 SHALL HAVE ±3% TOLARNACE MAXIMUM.
8. FOR USE WITH TEST SET ASM-464.
9. ALL GROUNDS SHALL BE TIED TO THE GROUNDING RINGS.

FIGURE 9. Firing Circuit Schematic

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4.6 Quality conformance failure. Failure classification shall be in accordance with MIL-STD-2074. Should a failure occur during testing under 4.5.3 and 4.5.4 the following actions shall be taken:

a. Immediately notify the procuring activity or designated representative.

b. Determine the cause and analyze the physics of the failure.

c. Determine if the failure is a recurring manufacturing problem or design deficiency.

d. Perform a failure analysis and determine appropriate corrective actions.

e. Prepare failure report, including proposed corrective actions and options (see 6.2.2).

f. Implement corrective action in all units for recurring manufacturing problems and design deficiencies, after validating the corrective action.

4.6.1 Validation of corrective action. After the occurrence of a failure, the test procedures for the individual testing shall be temporarily modified to check each power supply for previously noted nonconformance until it has been determined that satisfactory corrections have been incorporated. The procuring activity shall be notified of all modifications and corrections.

4.6.2 Restart of testing. Upon completion of repair after the occurrence of a failure, those tests where testing or retesting is to start shall be determined by the procuring activity. This procedure shall be repeated until all tests have been passed successfully to the satisfaction of the procuring activity.

4.7 Rejection and retest. A power supply which has been rejected during quality conformance inspection (See 4.5) may be reworked by the contractor or have parts replaced to correct defects and then be resubmitted. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector or procuring activity. Power supplies resubmitted and rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

4.8 Inspection of packaging. The inspection of the preservation, packing and container marking shall be in accordance with MIL-P-116.

5. PACKAGING

5.1 Packaging requirements. The power supply shall be preserved, packed, and marked as specified in MIL-P-116, MIL-STD-129 and MIL-STD-726 Code 4Q110000ZZZB4ZZZ.

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6. NOTES

6.1 Intended use. The power supply covered by this specification is intended for use in the Guided Missile Launcher LAU-7/A series (MIL-L-22620).

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Quantity of power supplies to be procured.
c. Responsibility for inspection (see 4.1).
d. First article inspection if applicable, invoking tests of 4.4.
e. Packing and packaging if different from 5.1.
f. Configuration control requirements invoking DOD-STD-480.

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.
3.2.1	Reliability Stress Analysis Report	UDI-R-21423
3.2.4	Program Parts Selection List	DI-MISC-80072
	Part Approval Request	DI-MISC-80071
3.3.1.2	Assignment of Serial Number Prefix Letters	DI-CMAN-80195
3.3.2.9	Failure Mode, Effects, and Criticality Report	DI-R-7085A
3.6	Electromagnetic Interference Control Plan	DI-EMCS-80199
3.7.2	Reliability Program Plan	DI-R-7079
	Reliability Mathematical Models	DI-R-7081
	Reliability Status Report	DI-R-7080
3.9	Engineering Drawings and Associated Lists	DI-E-7031
4.1.1	Quality Assurance Program Plan	UDI-R-21374A
4.4.1.1	Electromagnetic Interference Test Plan	DI-EMCS-80201
4.4.1.1, 4.5	Procedures, Test	UDI-T-21347
4.4.1.2	Electromagnetic Interference Test Report	DI-EMCS-80200
4.4.1.2, 4.4.5	Test Reports	DI-T-5247

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4.4.1.2, 4.4.5, 4.6	Report, Failure Summary and Analysis	DI-RELI-80255

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12L, AMDSL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No. 85576) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Air Systems Command, however information pertaining to qualification of products may be obtained from Commander Pacific Missile Test Center (Code 2043), Point Mugu, CA 93042.

6.4 Government-loaned property. The Government will loan the following property:

Item	Part Number
a. LAU-7 series Missile Launcher	58A164H891
b. LAU-7 Nitrogen Receiver	58A164D556

6.5 Definitions. Definitions are stated as follows:

a. Failure. A failure is defined as the inability of an item to perform within specified limits.

b. Repetitive failure. A repetitive failure shall be defined as the occurrence of two or more failures of the same part in identical or equivalent application by the same basic failure mechanism.

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject Term (Key Word) Listing.

Power Supply, Guided Missile Launcher
Power Supply, LAU-7
Power Supply, LAU-7, Guided Missile Launcher
Power Supply, Missile

Preparing Activity:
Navy-AS
(Project 1440-N074)

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